

Compressing quantum states

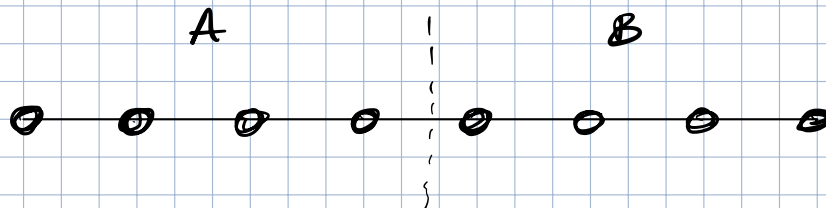
$$\dim H \sim d^L$$

L : # sites / level

d : local dimension

$d=2$ for $\uparrow \downarrow$

$$|\psi\rangle = \sum_{s_1 \dots s_L} C_{s_1 \dots s_L} |s_1 \dots s_L\rangle$$



$$|\psi\rangle = \sum_{ij} C_{ij} |i\rangle_A \otimes |j\rangle_B \quad \dim C = d_A \times d_B$$

Singular Value Decomposition (SVD)

$$C = U \Sigma V^\dagger$$

Isometries: $U^\dagger U = V^\dagger V = \mathbb{1}$

$$U U^\dagger \neq \mathbb{1} \quad V V^\dagger \neq \mathbb{1}$$

Σ : diagonal matrix with entries σ_k

$$\begin{aligned}
 |\psi\rangle &= \sum_{ijk} U_{ik} \sum_k V_{kj}^+ |i\rangle_A \otimes |j\rangle_B \\
 &= \sum_k \sigma_k \underbrace{\left(\sum_i U_{ik} |i\rangle_A \right)}_{|u_k\rangle_A} \otimes \underbrace{\left(\sum_j V_{kj}^+ |j\rangle_B \right)}_{|v_k\rangle_B}
 \end{aligned}$$

Schmidt decomposition

$$|\psi\rangle = \sum_k \sigma_k |u_k\rangle_A \otimes |v_k\rangle_B$$

$$\langle u_k | u_{k'} \rangle = \delta_{kk'}$$

$$\langle v_k | v_{k'} \rangle = \delta_{kk'}$$

$$\rho_A = \text{Tr}_B |\psi\rangle\langle\psi| = \sum_k \sigma_k^2 |u_k\rangle_A \langle u_k|_A$$

$$\rho_B = \text{Tr}_A |\psi\rangle\langle\psi| = \sum_k \sigma_k^2 |v_k\rangle_B \langle v_k|_B$$

Entanglement entropy:

$$\begin{aligned}
 S_A &= -\text{Tr} \rho_A \ln \rho_A = -\sum_k \sigma_k^2 \ln \sigma_k^2 \\
 &= S_B
 \end{aligned}$$

$$| \uparrow \uparrow \rangle = | \uparrow \uparrow \rangle = | \uparrow \rangle \otimes | \uparrow \rangle \quad \underline{\sigma_1 = 1}$$

$$S_A = -1 \ln 1 = \underline{0}$$

$$d_A = 2^{L_A}$$

$$d_B = 2^{L_B}$$

$$| \uparrow \uparrow \dots \uparrow \rangle = | \uparrow \dots \uparrow \rangle_A \otimes | \uparrow \dots \uparrow \rangle_B$$

$$\sigma_k \neq 0 \quad \text{for} \quad 1 \leq k \leq r$$

$$\sigma_k = 0 \quad \text{for} \quad r < k \leq \min(d_A, d_B)$$

The value of r tells you a lot about how entangled the system is.

$$\sigma_k = 1/\sqrt{r} \quad S_A = -\sum_k \frac{1}{r} \ln \frac{1}{r} = \sum_{k=1}^r \frac{1}{r} \ln r = \ln r$$

$\Rightarrow$ entanglement grows with r .

Slightly entangled states are well described by only a few Schmidt values.

Eckhart-Young: Keeping the χ largest σ_k 's is the best rank- χ

approximation to C , with discarded weight

$$E_\chi = \sum_{k > \chi} \sigma_k^2$$

States with little entanglement (only a few large Schmidt values / singular values) can be heavily compressed while keeping ϵ_x small.

Area law

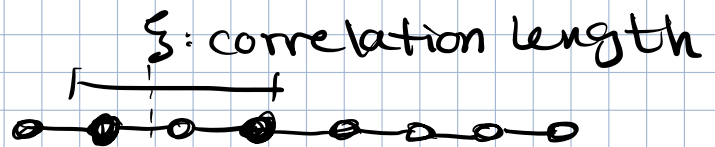
Ground state of a local, gapped Hamiltonian has

$$S_A \sim L^{D-1}$$

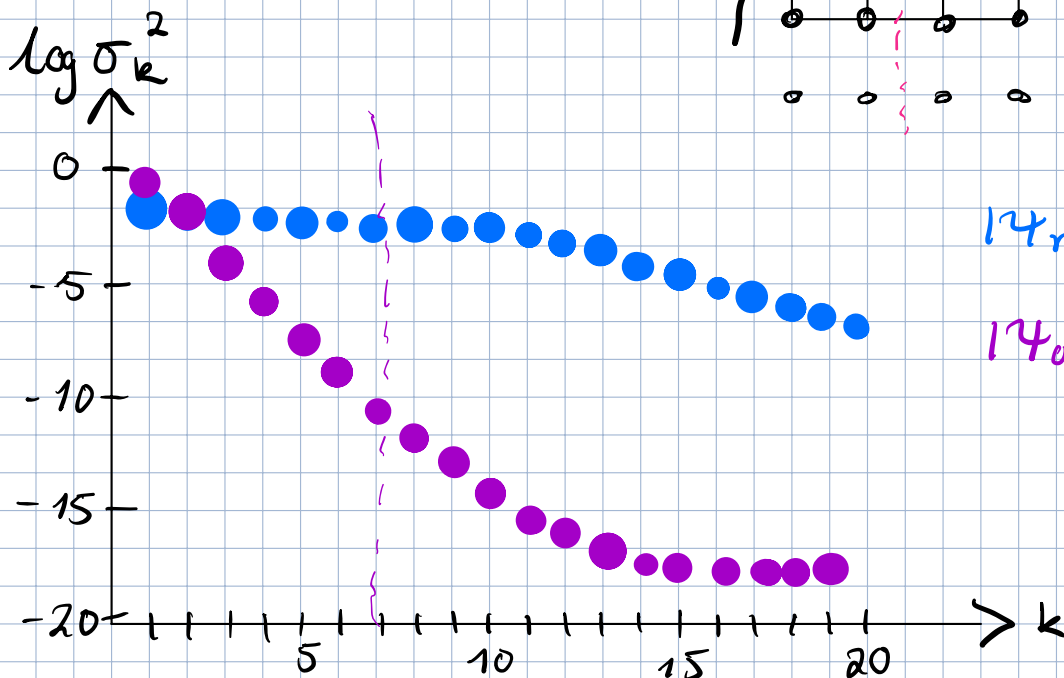
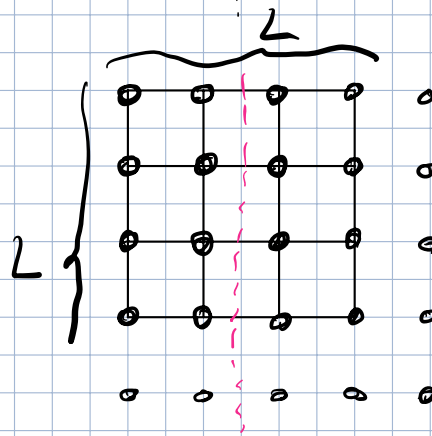
D : spatial dimension

(Volume law: $S_A \sim L^D$)

1D: $S_A = \text{const.}$



2D: $S_A \propto L$

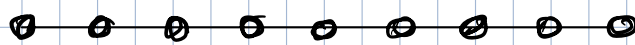


$| \psi_{\text{random}} \rangle$

$| \psi_0 \rangle$: ground state of a local gapped H.

Conclusion: Area Law states are well-suited for compression.
(Particularly in 1D)

Matrix Product States (MPS)

 1D chain of L sites

Can divide into A and B at any of the $L-1$ bonds.

MPS $\equiv$ Do an SVD on each of $L-1$ bonds.

General state:

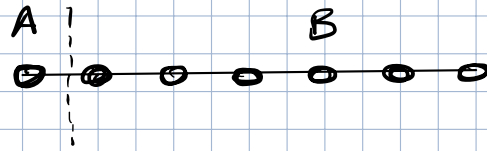
$$|\Psi\rangle = \sum_{s_1 \dots s_L} C_{s_1 \dots s_L} |s_1 \dots s_L\rangle$$

C : order- L tensor, d^L entries

Idea of MPS: rewrite C as the product of L matrices (one matrix per site)

Recipe to construct an MPS

Step 1:



* Cut after site 1.

* Reshape C into a $d \times d^{L-1}$ matrix $C_{s_1, (s_2 \dots s_L)}$

* Take SVD of C matrix.

(Schmidt decomposition of cut on bond 1.)

$$|\psi\rangle = \sum_{s_1, (s_2 \dots s_L)} C_{s_1, (s_2 \dots s_L)} |s_1\rangle \otimes |s_2 \dots s_L\rangle$$

$$C_{s_1, (s_2 \dots s_L)} = \sum_{a_1=1}^{r_1} U_{s_1, a_1} \sigma_{a_1} V_{a_1, (s_2 \dots s_L)}^+$$

$\swarrow$ site 1
 $A_{a_1}^{[1] s_1} \equiv U_{s_1, a_1}$ Matrix $A^{[1] s_1}$ rows of U
 (vector)

$$C_{s_1, (s_2 \dots s_L)} = \sum_{a_1=1}^{r_1} A_{a_1}^{[1] s_1} \underbrace{\tilde{C}_{a_1, (s_2 \dots s_L)}}_{\sigma_{a_1} V_{a_1, (s_2 \dots s_L)}^+}$$

Step 2:



* Cut on bond 2

* Reshape $\tilde{C}$ into an $(r_1 d) \times d^{L-2}$ matrix

(Before reshape, $\dim \tilde{C} = r_1 \times d^{L-1}$)

* Find SVD of $\tilde{C}$

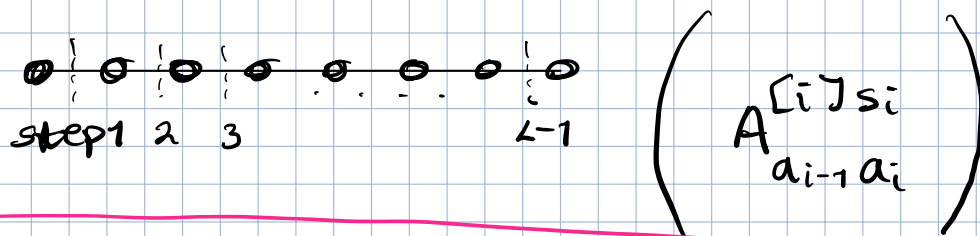
$$\tilde{C}_{(a_1 s_2), (s_3 \dots s_L)} = \sum_{a_2=1}^{r_2} U_{(a_1 s_2), a_2} \sigma_{a_2} V_{a_2, (s_3 \dots s_L)}^+$$

$$A_{a_1 a_2}^{[2] s_2} = U_{(a_1 s_2), a_2} \quad \tilde{C}_{a_2, (s_3 \dots s_L)} = \sum_{a_2=1}^{r_2} A_{a_1 a_2}^{[2] s_2} \tilde{C}_{a_2, (s_3 \dots s_L)}$$

$$C_{s_1 \dots s_L} = \sum_{a_1=1}^{r_1} A_{a_1}^{[1] s_1} \tilde{C}_{a_1, (s_2 \dots s_L)}$$

$$= \sum_{a_1=1}^{r_1} \sum_{a_2=1}^{r_2} A_{a_1}^{[1] s_1} A_{a_1 a_2}^{[2] s_2} \tilde{C}_{a_2, (s_3 \dots s_L)}$$

Continue for $L-1$ steps in total



Order- L tensor

$$C_{s_1 s_2 \dots s_L} = \sum_{a_1 a_2 \dots a_{L-1}} A_{a_1}^{[1] s_1} A_{a_1 a_2}^{[2] s_2} \dots A_{a_{L-2} a_{L-1}}^{[L-1] s_{L-1}} A_{a_{L-1}}^{[L] s_L}$$

$$= A^{[1] s_1} A^{[2] s_2} \dots A^{[L-1] s_{L-1}} A^{[L] s_L}$$

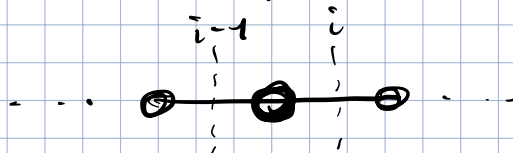
Product of L matrices

$$|\psi\rangle = \sum_{s_1 \dots s_L} A^{[1]s_1} \dots A^{[L]s_L} |s_1 \dots s_L\rangle$$

Matrix product state

Properties:

- * One set of d matrices $A^{[i]s_i}$ per site $s_i = 1, \dots, d$
- * The dimension of $A^{[i]s_i}$ is $r_{i-1} \times r_i$



#Schmidt values

Dim $A^{[i]}$ depends on entanglement when cutting bond $i-1$ and bond i .

- * $A^{[1]s_1}$: row vector
 - * $A^{[L]s_L}$: column vector
- } Important for $C_{s_1 \dots s_L}$ to be a number!